

Exam 5 marks (scenario based)

Network devices26/Nov/11
(diagram)Organisation of Network

• End user devices & Network Devices.

• Connecting Devices.

• Five categories of Network Devices

• Repeater * (discussed in detail)

• Functions of repeater

Attenuation = signal weak.

Token (jis se pas taken hoga wahi data send)

• HUB

Types of hub

• what is a bridge

> switch

> 2 Layer switches

> 3 layer switch

> Routers

> Routing Principle

> Differences

> Gateway

A

B

C

D

P

gateway

Router

Bridge

Repeater

hub

Passive

4/Dec/24

Reliable data transfer

3-44, 3-45, 3-46, 3-47, 3-48, 3-50, 3-52

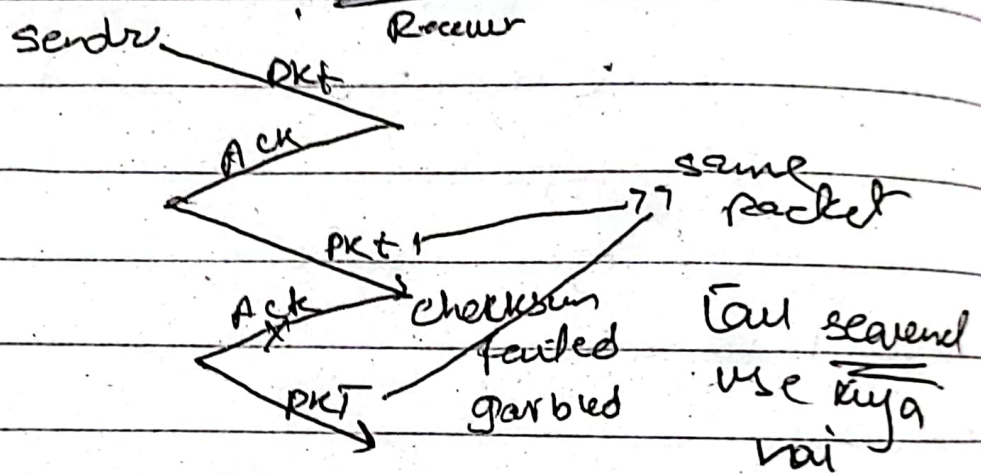
3-48
Exam
Q How you will develop reliable data transfer

USE checksum
in ACK & NAK

Timeout & Retransmit
Sequence Number

3-55 3-56 (Draw)

FSM



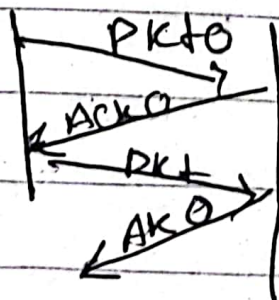
Q.1 3-58

During sequence use 0 & 1

Ans we don't need

• Discard duplicate

3-59 Q.2 NAK



3.0

3-61

packet send krne ke baad timer chlo g
hai take reanalyse aurand wait krni
ke bad pkt dobara send krni.

3-64

3-65 Exam

rdt 3.0

Performance X

- | | | |
|-----|--|---|
| 1.0 | reliable channel | 0 |
| 2.0 | checksum & Ack NAK | 0 |
| 2.1 | Sequencing | 1 |
| 2.2 | retransmit NAK any Ack
if corrupt pkt deliv
receiver send duplicate
& again Ack will send | 1 |
| 3.0 | Timeout | 1 |

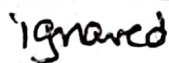
17/12/20

pipeline

missouri

$n = 3$

Winda
Stück
PKT. wasser
deiner



Time out

ACKPT

~~RKfay~~

~~PK 10/11~~

~~PKA~~

4567789

discard

0 1 2 3 4 5 6 7 8 9

selective repeat

~~0 1 2~~

~~0 1 2 3~~

~~0 1 2 3 4~~

0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9

0 1 2 3

0 1 2 3 4

0 1 2 3 4 5

0 1 2 3 4 5 6

0 1 2 3 4 5 6

0 1 2 3 4 5 6

send pkt#

0 1 2 3 4 5 6 7

0 1 2 3 4 5 6 7 8

0 1 2 3

0 1 2 3 4

0 1 2 3 4 5

0 1 2 3 4 5 6

0 1 2 3 4 5 6

0 1 2 3 4 5 6

5 receive
buffer

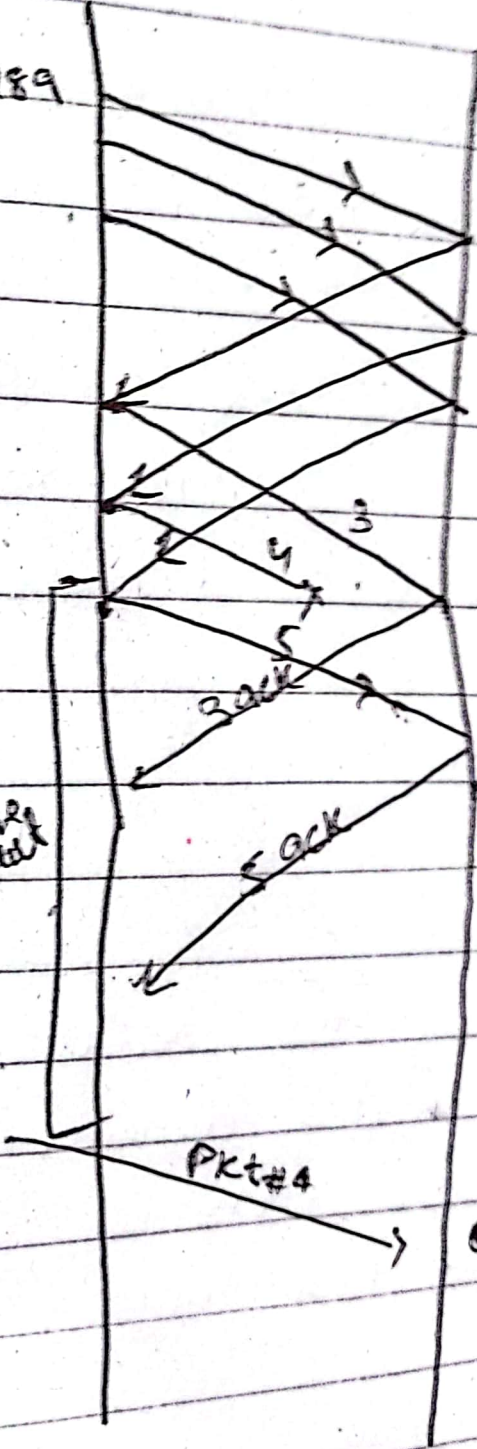
0 1 2 3 4 5 6 7 8

Time out

Pkt#4

3ack

5ack



24/12/24

TUE

Chapter #3

41 to 108

Slides

~~40~~ included

Chapter #4

Slide 4-4

→ network layer sab mai hata hai
edges mai or core devices

Exam → what are two key network functions 4-5

Ans forwarding & routing

Routing algorithm → deciding best path
4-5 discussed in detail

4-6

4-7

Routing algo decide which is the best path, it's dynamic

→ header dekh ke agla aut^{put} router decide hata hai

100 --	2
101100	3

→ all routers communicate with each other

→ central plane (global decision)

→ data plane

Trade fair algo has local forwarding table just network manager
khud banate hai 10100 3 1/2
now they're campus has 2 approx

4-8 not much discussed

SDN (Software defined networking)

→ Router can't communicate each other

→ it's not dynamic

4-9 not much discussed

4-10 mel u n

~~4-11 n n n~~

4-12 Discussed

Q Why widely adapted is internet?
Simplicity, replicat,

4-14 Switching fabric

11-14 Section
 Pocket Auftragen auf dusri tara daga
 brain

→ rawing processor is a brain

4-16

match plus action

4-17

4-18

4-18
4-19 Longest prefix matching

4-23

4-24

Switching fabric

4-25

4-26 Via memory

4-27 Bus

4-28 Crossbar

4-29

4-30 Hol. discussed

2 mark ques.

✓ Improved
4-31

Drop kisko kma
hai

4-34 tail drop
marking

4-35

4-36

-37

-38

not discussed
much

Ip will be discussed

WEP
The

mai class

miss hai hai

2/4/25

4-65

NAT

zoroni

Exam \$ 10.0.0/24

Class full

IP Address

Difference
Classless CIDR

A 1248

B 14616

C 1824

D

24 + 8
Subnet host

10.0.0.1
Subnet host

There're 3 organisations & there are
1500 devices

Why using Classless CIDR?

What's an issue with classfull

Explain with example?

Ans



300 devices

There're 3 organisation Org 1 have 1000 devices

Org 2 has 500 devices

Org 3 has 500 devices

miss IP hai de saku hai n

2

2

NAT summary

4-66

4-67

4-68

4-69 NAT is controversial?

Source IP, port

New Port no., NAT IP address

4-70

4-71

} not much discussed

IPV6 vs IPV4 what's missing?

Q compare IPV6 & IPV4 Datagram

4-43

4-73

4-75

Tunneling

See water paint missing

kanu exam nae aya

Chapter # 5

7/11/25

5-4 network layer function

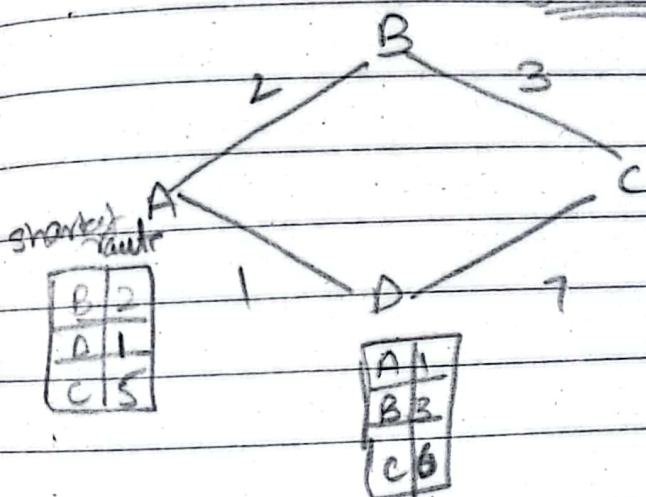
5-5, 5-6 Per-router central plane vs ~~SDN~~ SDN

5-9 Routing protocol

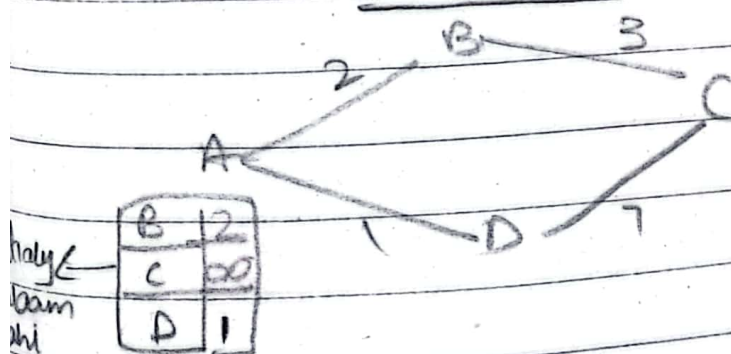
good path: least cost, fast, least congestion

5-10 graph abstraction link costs

5-11 Routing algorithm classification
with explained global

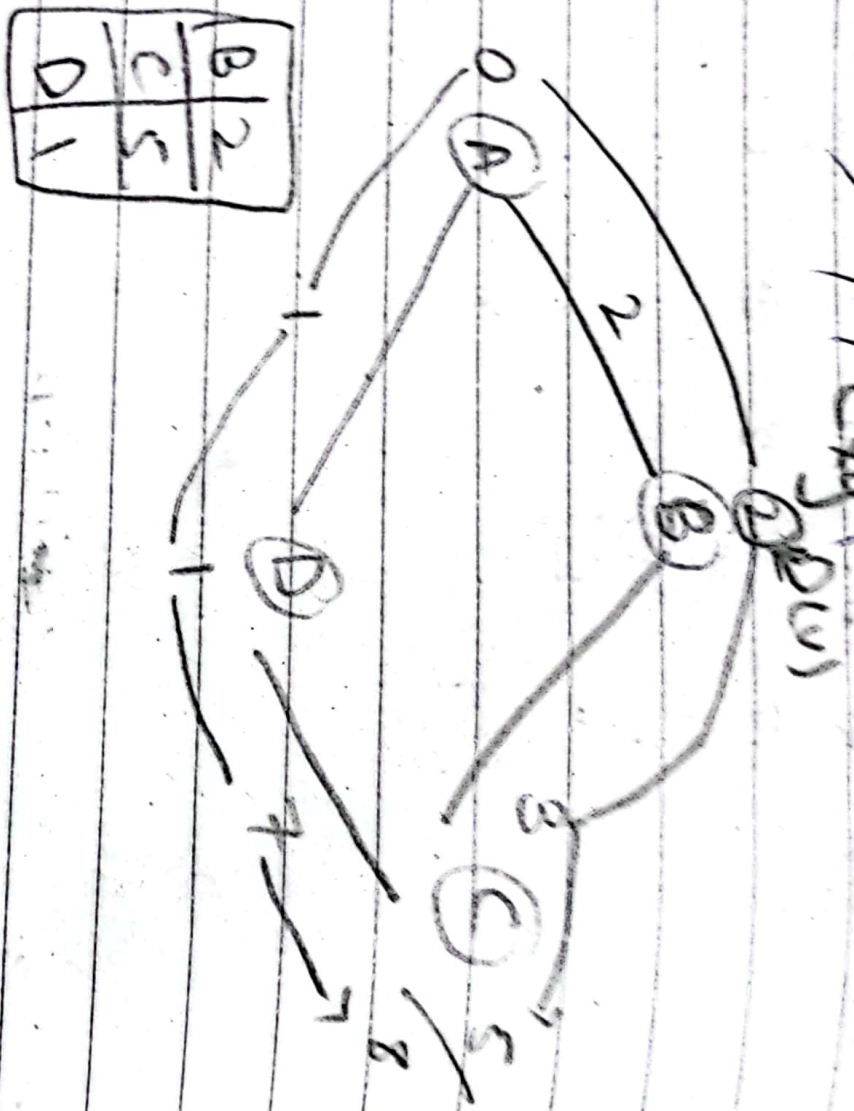


Decentralised

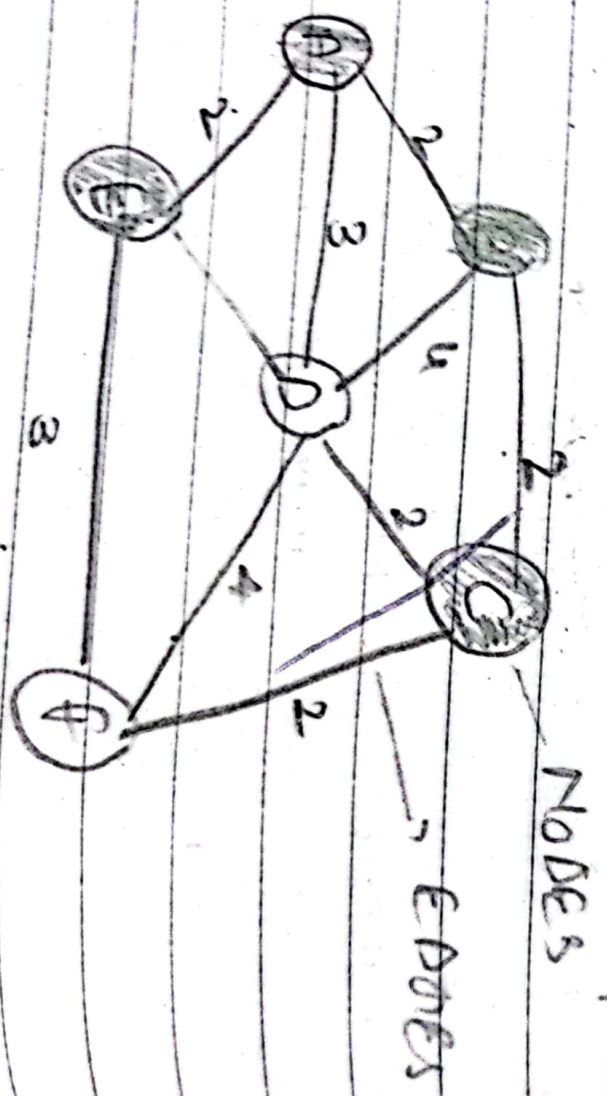


Directly connected
network ka matlab hai

5.12 Dijkstra's Shortest Path Algorithm



Dijkstra's algorithm: an example



To mark page no
Nauri Ekharigan

Marked

2

66

3

3

2

Yours 6
near library
Key 653

no direct

5

15

5

1

5

3

5

19

Case A-A 20

24/11

A-B-22

4

A-C 24

CD 201

A-D 23

DA

A-E 22

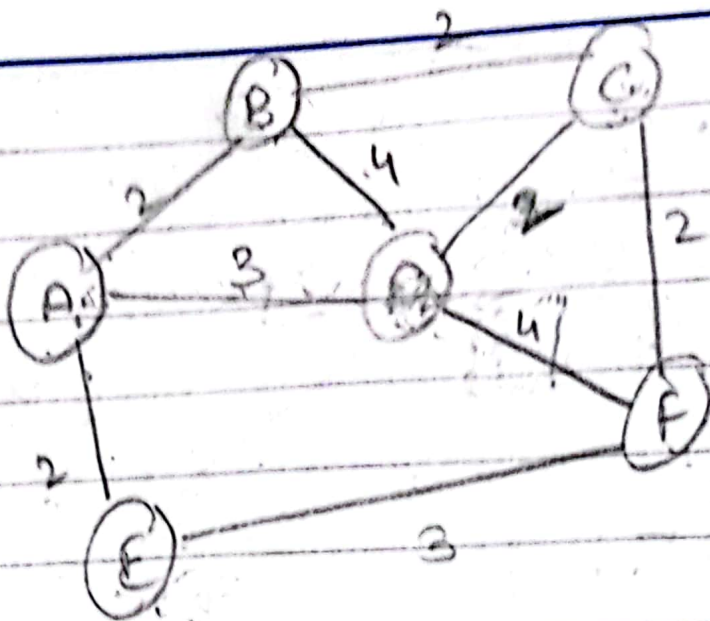
77

AF 25

$$A \subseteq B \subseteq C \subseteq D$$

Find shortest path $\frac{1}{2}$ cost
from A to all other nodes in G.
The network

1103AB



marked	A	B	C	D	E	F
D	∞	∞	∞	0	∞	∞
A	3	4	2		∞	4
C	3	4			∞	4
A	3	4			5	4
B					5	4
F					5	
E					5	

Cost

$$D-A = 2$$

$$D-B = 4$$

$$D-C = 3$$

$$D-D = 0$$

$$D-E = 5$$

$$D-F = 5$$

Path

$$A \leftarrow D$$

$$B \leftarrow D$$

$$C \leftarrow D$$

$$D \leftarrow D$$

$$E \leftarrow C \leftarrow D$$

$$F \leftarrow D$$

D	A	B	C	D	E	F
D	∞	∞	∞	0	∞	∞
C	3	4	2		∞	4
A	3	4			∞	4
B		4			5	4
E					5	

Path

Cost

$E \leftarrow A \leftarrow D = 5$

$F \leftarrow D = 4$

$B \leftarrow D = 3$

$A \leftarrow D = 2$

$C \leftarrow D = 0$

$D \leftarrow D = 0$

8/Jan/25

S-31

Distance vector

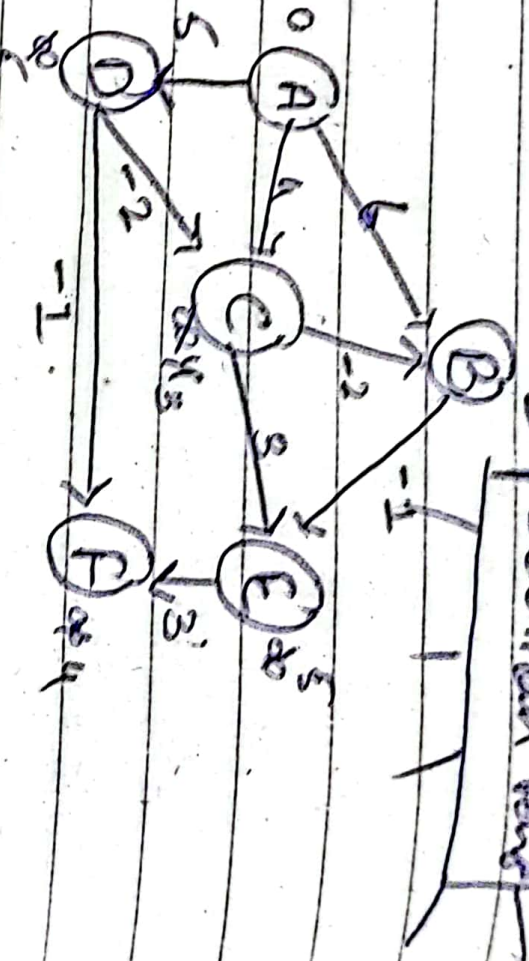
S-32

no need for formula learn

concept

2 (E, S)

Bellman ford



(A,B) (A,C) (A,D) (B,E) (C,D) (C,E)
(D,C) (D,F) (E,E)

Iteration = 9 - 1

1st iteration
concept

1st iteration =

now second
iteration

(A,B) (A,C) (A,D) (B,E) (C,D) (C,E)
(D,C) (D,F) (E,E)

iteration hai
Same hai
jaisa hai

cast from ft

$B = 1$

$C = 3$

$D = 5$

$E = 0$

$F = 8$

Path

$A \rightarrow D \rightarrow C \rightarrow B$

$A \rightarrow B \rightarrow D \rightarrow C$

$A \rightarrow C \rightarrow$

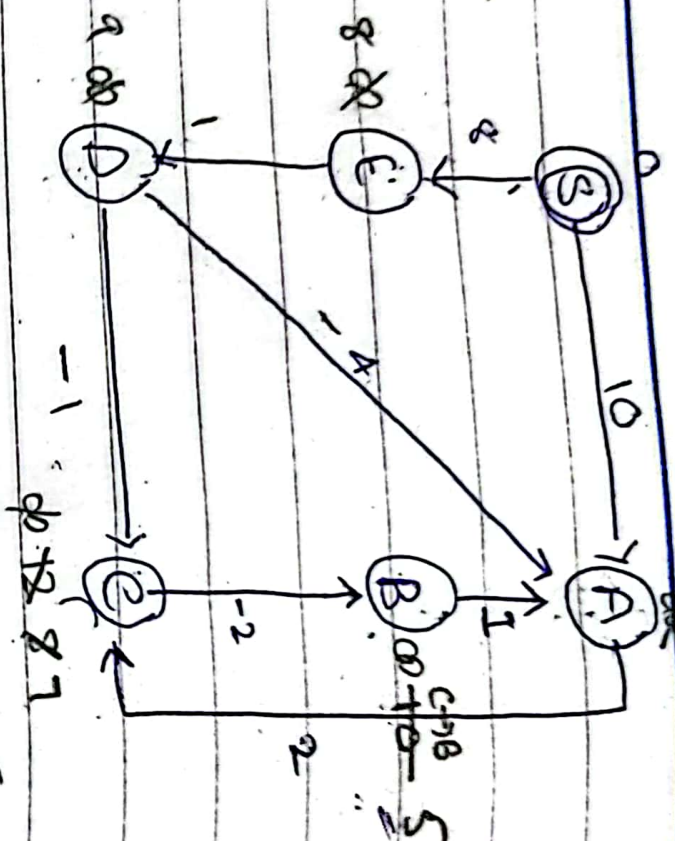
$A \rightarrow D \rightarrow A \rightarrow D$

$A \rightarrow E \rightarrow$

$A \rightarrow D \rightarrow C \rightarrow B \rightarrow E$

$A \rightarrow F \rightarrow A \rightarrow E \rightarrow B \rightarrow E$

$A \rightarrow D \rightarrow C \rightarrow B \rightarrow E \rightarrow F$



(S, ϵ) (S, A) (D, A) (D, C) (A, C) (B, A)
 (C, B) (C, D)

8-127

1st iteration

2nd iteration

3rd iteration

cases

- A_2 $S \rightarrow A$ $S \rightarrow E \rightarrow D \rightarrow A$
- B_2 $S \rightarrow B$ $S \rightarrow C \rightarrow D \rightarrow A \rightarrow C$
- C_2 $S \rightarrow C$ $S \rightarrow A \rightarrow C$
- D_2 $S \rightarrow D$ $S \rightarrow E \rightarrow D$
- E_2 $S \rightarrow E$ $S \rightarrow E$

Diagnostik

Subnet

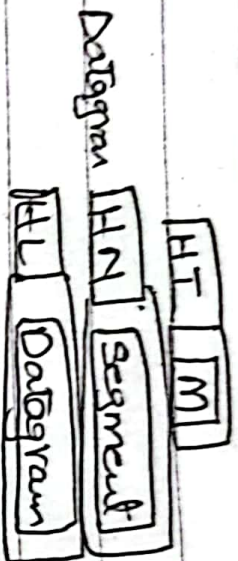
ei dr

Dijkstra

Bellevue

Chapter #6

6-4 link layer introduction



6-5 link layer context.

6-6 link layer, services.

6-7

6-8 where is the link layer implemented

6-9 not much discussed

6-11 Error detection

6-12 Parity checking

Two dimensional

6-13

6-14 Cyclic redundancy

$$\text{Key } x^3 + x^2 + 1 \rightarrow x^3 + x^2 + x^1 + x^0$$

1 1 0 1

$$x^5 + x^2 \rightarrow \text{Key}$$

$$x^3 \quad x^4 \quad x^1 \quad x^0$$

$$1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad \text{Data}$$

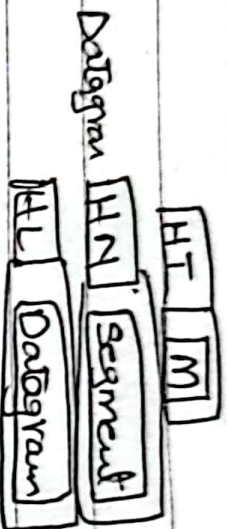
crc bits

$$\begin{array}{r} 1101 \quad 10010000 \quad \text{Data} \\ 1101 \quad \underline{\quad\quad\quad} \\ 01000 \end{array}$$

01000

Chapter #6

6-4 link layer introduction



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6-7

6-8 where is the link layer implemented

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6-11 Error detection

6-12 Parity checking

two dimensional

6-13

6-14 cyclic redundancy

$$\text{Key } x^3 + x^2 + 1 \rightarrow x^3 + x^2 + x^1 + x^0$$

1101

$$x^5 + x^2 \rightarrow \text{key}$$

$$1 \quad x^3 \quad x^4 \quad x^1 \quad x^0$$

Data

$$\begin{array}{r} \text{Key } 1101 \quad \text{Data } 10010000 \\ \hline 1101 \\ \hline 01000 \end{array}$$

crc bits

01000

zaman kısıtlı

1111

cccc

1101 1001 0000 0000

1101 ↓

1000

1101 ↓

1010

1101 ↓

1110

1101 ↓

111000

1101

001

1101

$$\text{Key} = x^4 + x^3 + x^2 + 1$$

$$111001$$

$$\text{Data} = x^7 + x^6 + x^3 + x^2$$

$$110011010000$$

$$\begin{array}{r} 11001 \overline{) 110011010000} \\ \underline{1100111111} \\ 0000010100 \\ \underline{110011} \\ 011010 \\ \underline{11001} \\ 0110 \end{array}$$

Da

$$\begin{array}{r} 11001 \overline{) 11001101010} \\ \underline{11001} \\ 10101 \\ \underline{11001} \\ 11001 \\ \underline{11001} \\ 00000 \end{array}$$

CRC

Date 10.11.2011

Chapter #7

14/11/25

imp

7-2

Hand off

7-5, 7-6, 7-7, 7, 8, 7.9, 7-10

7-13, 7-22, 7-23, 7-24, 7-25

7-26 7-27

↓ how to avoid confusion

~~XXXX~~

Chapter 4

data plan
= $\bigotimes \bigotimes \bigotimes \bigotimes$

control plane

routing algo has local forwarding table through which it's decided datagram will be sent to which output

11010101

$\bigotimes \bigotimes \bigotimes \bigotimes$

Now local forwarding table can be compute by time way

Traditional

each router algo communicate with other routing algo

$\bigotimes \bigotimes$

SDN (Software defining)

networking

central controller computes instead forwarding entries in router.

router don't communicate with each other.

Why industry adopted is internet?

• Simplicity of mechanism has ~~attained~~

• Sufficient provision of bandwidth

allows performance of real-time

applications to be good enough for most of the time.

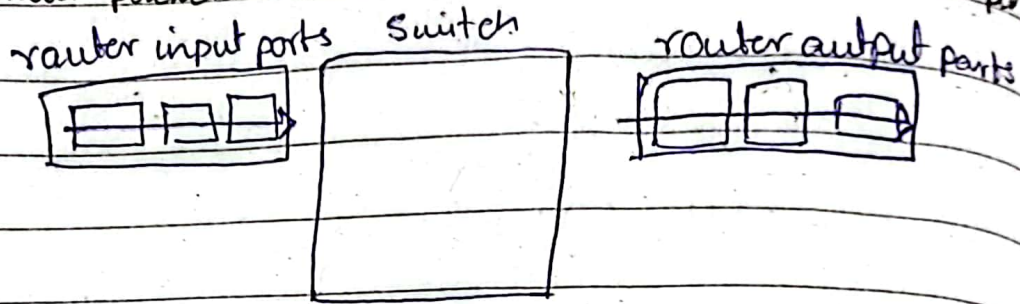
• replication, application-layer distributed service

(data centres, content) connecting class B user allow

services to be provided from multiple hosts

Q Inside Router

How packets move from input part to output part

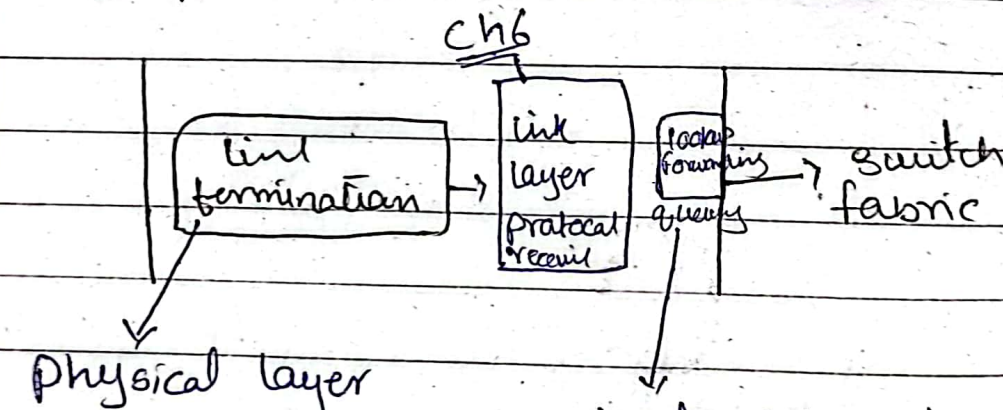


• Packets move from input part to output part using switching fabric (heart of the router)

(consist routing processor)

- Performs control plane function
- Controls switch fabric.
- install forwarding tables at input part

Input Port function



Physical layer responsible for receiving bit and transmission over physical layer

decentralised switching:

- using header field values, lookup output part using forwarding table in input part memory (match and switch)

• input part queue

allowing faster than receiving rate into switch fabric.

forwarding

destination based
forward based on any
destination IP address
(traditional)

Generalised Forwarding
forward based
on any set
of header field values

Destination based forwarding

* Packet output is determined by longest prefix matching.

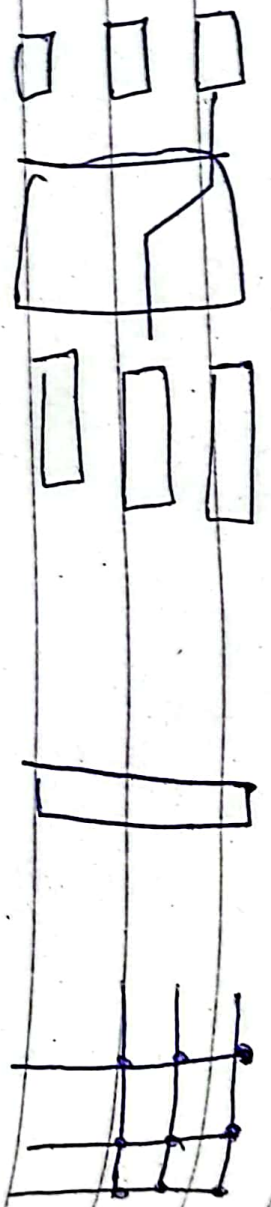
Switching fabric

transfer packet from input to
appropriate output link determined
by longest prefix matching.

Switching rate rate at which
packets are transfer from input to
output

Types of Switching fabric

Switching via memory
~ ~ ~ Bus



interconnection

input part memory output part
data transfer